

Task I: Read the following abstracts. Identify and write 2 objective/purpose statements (in your own words) mentioned implicitly or explicitly in the abstracts given.

1. To ensure the safety of autonomous Multi-rotor UAVs flying in urban airspace, they should be capable of avoiding collisions with unpredictable dynamic obstacles, such as birds. UAVs must consider both relative position and relative velocity to avoid moving obstacles. Model predictive control (MPC) can consider the multiple collision avoidance constraints in a constrained optimization framework. This study proposes a chance-constraints based on obstacle velocity (CCOV) method, which can be combined with previous positional chance constraint methods to account for uncertainty in both position and velocity. This effectively prevents collision with high-velocity obstacles, even in a noisy environment. The proposed method has been performed on a numerical simulation built in MATLAB.

2. Eating is an essential activity of daily living (ADL) for staying healthy and living at home independently. Although numerous assistive devices have been introduced, many people with disabilities are still restricted from independent eating due to the devices' physical or perceptual limitations. In this work, we present a new meal-assistance system and evaluations of this system with people with motor impairments. We also discuss learned lessons and design insights based on the evaluations. The meal-assistance system uses a general purpose mobile manipulator, a Willow Garage PR2, which has the potential to serve as a versatile form of assistive technology. Our active feeding framework enables the robot to autonomously deliver food to the user's mouth, reducing the need for head movement by the user. The user interface, visually-guided behaviors, and safety tools allow people with severe motor impairments to successfully use the system. We evaluated our system with a total of 10 able-bodied participants and 9 participants with motor impairments. Both groups of participants successfully ate various foods using the system and reported high rates of success for the system's autonomous behaviors. In general, participants who operated the system reported that it was comfortable, safe, and easy-to-use.

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| 1. To develop a UAV collision-avoidance method that considers both position and velocity of moving obstacles to improve flight safety. |
| 2. To design and evaluate a robotic meal-assistance system that helps people with motor impairments eat independently. |

Task II: Read the following background and develop a problem statement for it. (Word count: 35-50)

In recent years, technology has played a significant role in shaping various aspects of our lives, including education. The widespread adoption of online learning platforms, such as Coursera and Udemy, has allowed learners to access high quality educational content from the comfort of their homes. While online learning offers several benefits to learners, it is also associated with some challenges, such as a lack of personal interaction and guidance. To address these challenges educators have started exploring the potential of Artificial Intelligence (AI) in online learning.

Despite the growing popularity of online learning platforms, students often face difficulties due to limited personal interaction and lack of individualized guidance. This reduces engagement and learning effectiveness. Therefore, there is a need to explore how AI can address these gaps by providing personalized support and interactive learning experiences.
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